

Comprehensive Corrected Notes

From the single-cell birth–death–catastrophe process
to a burst-aware Basic Model of Viral Replication

Working notes for a thesis chapter and a paper
(all corrections applied; every major item in this folder represented)

Compiled August 5, 2026; mathematics verified independently

Abstract

These notes consolidate **every major result** in the folder 4 BDC additional and BMVR (and the co-located BDC core), with **all known mathematical corrections applied**. They are written so that a thesis chapter and a paper can be drafted from them without re-deriving or re-hunting errata.

Tags: **[CORRECTED]** replaces a thesis error; **[NEW]** is present in spirit in the sources but stated cleanly here (or found on re-checking); **[CAVEAT]** is a modelling subtlety; **[OPEN]** needs more work; **[SOURCE]** points to the origin path (from folder root 4 BDC additional and BMVR/).

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1 How to use these notes

1. **Thesis chapter.** Sections 4–7 and 9–11 form the single-cell theory chapter. Sections 20–26 form the population / BMVR chapter. Section 34 is a short honest contrast, not the main claim.
2. **Paper.** The publishable core is: corrected single-cell closed forms + burst size = QSD + renewal BMVR with closed kernels + $p_{\text{eff}}(r)$ + invariant R_0 + flooding theorem (with $L = 1$ law match) + r -trade-off.
3. **Do not claim for HIV** the pure one-type BDC terminal catastrophe; see Section 34.
4. **Figures already generated:** `../figures/kernels.pdf`, `overlay_V.pdf`, `overlay_I.pdf`, `overlay_rel_diff.pdf`, `overlay_growth_phase.pdf`, `overlay_V_with_naive.pdf` (script: `../figures/plot_bmvr_comparison.py`).
5. **Result 20.1 verification:** Section 21; script `verify_result_20_1.py` (this folder); **53/53 PASS**.

Paths below are relative to the folder root 4 BDC additional and BMVR/. This document lives in `new_notes3/`.

2 Map of materials in this folder

Path (from folder root)	Content
3 BDC core/document MAIN/sections/	Ch.3 sources: process, moments I, J, K, V , variance (contains errors; corrected in this document)
3 BDC core/Chapter3_BDC_definition_main_results.pdf	Compiled Ch.3 snapshot
document MAIN/sections/	Ch.4 sources: PGF PDEs, p_n , QSD, burst size, BMVR attempt §9 (replaced here)
Chapter4_BDC_further_results.pdf	Compiled Ch.4 snapshot
new_notes/BMVR_extension_notes.tex	Working notes: consolidation, renewal BMVR, flooding, Part IV errata catalogue
new_notes2/Core_Publishable_Work_Summary.tex	Earlier clean summary: overlays, HIV, reset, Cases 1–5, boolean partial release
new_notes3/Comprehensive_Corrected_Notes.tex	This document (corrected master notes)
new_notes3/verify_result_20_1.py	Numerical suite for Result 20.1
new_notes3/verify_result_20_1_report.txt	Machine log (53/53 PASS)
new_notes3/verify_figures/	Verification plots (D, E, F, H)
N=2 Immediate Transfer content/N_eq_2.tex	Chained immediate-transfer model; rupture size laws
figures/	Kernel and NEW vs CLASSICAL BMVR overlays; <code>plot_bmvr_comparison.py</code>
paper.pdf	Compiled paper snapshot

Source-tag convention. Inline [SOURCE] tags cite paths from the folder root. Prefer reading `new_notes3/` (this file) and `new_notes2/` for cleaned statements; raw chapter `.tex` files remain the historical source of errors listed in Part I.

3 Canonical notation (use everywhere)

Symbol	Meaning	Notes / collisions to avoid
β, μ, δ	birth, death, catastrophe (per capita)	thesis also uses λ for birth — drop λ
X_t	intracellular count (CTMC)	
R	absorbing rupture state	thesis Ch.3 uses H — use R only
W_t	released count from one cell	0 until burst; was also Y_t
τ	burst time (may be ∞)	
$\mathcal{K}$	burst size (random)	never use bare K for this
$I(t)$	$\mathbb{P}\{\text{no burst by } t\}$	not infected-cell count
$D(t)$	$\mathbb{P}\{\text{cleared internally, no burst}\}$	$= p_0(t)$ in Ch.4
$\hat{I}(t)$	still productively infected $= I - D$	thesis formula wrong for $\mu > 0$
$J(t), K(t)$	$\mathbb{E}[X_t], \mathbb{E}[X_t^2]$	K is second moment only
$V(t)$	$\mathbb{E}[W_t]$ cumulative mean release	not free-virion count
a, b	characteristic roots, $b < 1 < a$	
A, B	$a - 1, 1 - b$	simplify almost everything
θ	$\beta(a - b)$	growth exponent
T	target-cell count (held fixed)	BMVR
γ	infection rate	not β
$\mathcal{I}, \mathcal{V}$	infected cells, free virions	BMVR population counts
c	virion clearance	BMVR
$d_{\mathcal{I}}$	infected-cell removal (phenomenological)	not δ
p	constant release rate (classical BMVR)	
$p_{\text{eff}}(r)$	effective release rate at growth rate r	
$g(a)$	release kernel (expected flux at age a)	
$S(a)$	survival / productive kernel	$\hat{I}$ for BDC
q	$\gamma T / (\gamma T + c)$ infection success (branching)	
L	$a(1 - b) = a - \mu/\beta$ flooding parameter	

Part A — Single-cell BDC (corrected)

4 Process definition

[SOURCE] 3 BDC core/document MAIN/sections/04_process_definition.tex; corrected statement also in new_notes2/Core_Publishable_Work_Summary.tex §“The single-cell process”, new_notes/BMVR_extension_notes.tex Part I.

Starting from $X_t = 1$ (one founder) unless stated, the process jumps

$$\begin{array}{ll}
 n \rightarrow n + 1 & \text{at rate } \beta n, \\
 n \rightarrow n - 1 & \text{at rate } \mu n, \\
 n \rightarrow R & \text{at rate } \delta n.
 \end{array}$$

On the jump to R the cell bursts: W_t is set to the pre-burst load and stays there. Classical references: Brockwell–Gani–Resnick; Karlin–Tavaré. What matters for applications is the closed-form family below.

[SOURCE] joint process (X_t, W_t) : same file 04_process_definition.tex (“Post catastrophe count”); before τ , $W_t = 0$; at catastrophe, $W_t \leftarrow X_t$ and $X_t \leftarrow R$.

5 Root algebra

[SOURCE] 3 BDC core/document MAIN/sections/03_reference_values.tex, 05_analytical_quantities.tex
 A, B notation and $AB = \delta/\beta$ emphasised in new_notes/BMVR_extension_notes.tex Part I.

$$\eta = \frac{\beta + \mu + \delta}{2\beta}, \quad a = \eta + \sqrt{\eta^2 - \frac{\mu}{\beta}}, \quad b = \eta - \sqrt{\eta^2 - \frac{\mu}{\beta}}. \quad (1)$$

Write $A := a - 1 > 0$, $B := 1 - b > 0$, $\theta := \beta(a - b)$.

Result 5.1 (Vieta and the workhorse identity).

$$ab = \frac{\mu}{\beta}, \quad a + b = \frac{\beta + \mu + \delta}{\beta}, \quad AB = \frac{\delta}{\beta}, \quad A + B = a - b. \quad (2)$$

In particular $b < 1 < a$ always, and $\beta(1 - a)(1 - b) = -\delta$. Also $\theta = \sqrt{(\beta + \mu + \delta)^2 - 4\beta\mu}$.

[NEW] Identity $AB = \delta/\beta$ is elementary and should sit in every reference table; it cancels prefactors throughout Ch.3.

6 Three fixation functions

[SOURCE] I in Ch.3 05_analytical_quantities.tex; $D = p_0$ in document MAIN/sections/05_state_probabilities.tex
 correct $\hat{I} = I - D$ in new_notes/BMVR_extension_notes.tex Part I (thesis $\hat{I}$ formula wrong for $\mu > 0$).

Definition 6.1.

$$I(t) = \mathbb{P}\{\text{no burst by } t\}, \quad (3)$$

$$D(t) = \mathbb{P}\{\text{internal extinction by } t, \text{ no burst}\}, \quad (4)$$

$$\hat{I}(t) = \mathbb{P}\{X_t \notin \{0, R\}\} = I(t) - D(t). \quad (5)$$

Both I and D satisfy the Riccati equation

$$f' = \beta(f - a)(f - b) = \mu - (\beta + \mu + \delta)f + \beta f^2, \quad (6)$$

with $I(0) = 1$ and $D(0) = 0$. Writing $w := e^{\theta t} \geq 1$:

Result 6.2 (Closed forms for I , D , $\hat{I}$).

$$I(t) = \frac{aB + bAw}{B + Aw}, \quad (7)$$

$$D(t) = \frac{ab(w - 1)}{aw - b}, \quad (8)$$

$$\hat{I}(t) = \frac{(a - b)^2 w}{(B + Aw)(aw - b)}. \quad (9)$$

Limits: $I(\infty) = D(\infty) = b$, $\hat{I}(\infty) = 0$, $\hat{I}(0) = 1$, $\hat{I}'(0) = -(\mu + \delta)$.

Remark 6.3 ([CORRECTED] thesis $\hat{I}$ formula). Ch.3 gives $\hat{I}(t) = \hat{\eta}/(1 - (1 - \hat{\eta})e^{\hat{\eta}\beta t})$ with $\hat{\eta} = (\beta + \mu + \delta)/\beta$, from the wrong ODE $\hat{I}' = -(\beta + \mu + \delta)\hat{I} + \beta\hat{I}^2$. The branching term on a birth must be $I^2 - D^2$, not $\hat{I}^2$. Error is invisible when $\mu = 0$ ($D \equiv 0$). At $(\beta, \mu, \delta) = (1, 0.2, 0.05)$, $t = 1$: correct $\hat{I} \approx 0.8088$, thesis formula ≈ 0.6675 . Ch.4 state probabilities p_n sum over $n \geq 1$ to the correct $\hat{I}$.

Result 6.4 ([CORRECTED] multiplicity of infection — survival). *For $X_0 = k$ independent founders (one cell, linear rates, shared catastrophe):*

$$I_k(t) = I(t)^k, \quad D_k(t) = D(t)^k, \quad \hat{I}_k(t) = I(t)^k - D(t)^k. \quad (10)$$

Thesis claim $\hat{I}_k = \hat{I}^k$ is false.

7 Moments and mean release

[SOURCE] 3 BDC core/document MAIN/sections/05_analytical_quantities.tex, 07_expected_number_08_variance_and_standard_deviation_of.tex (and duplicate 10...); simplifications/corrections in new_notes/BMVR_extension_notes.tex Part I, new_notes2/Core_Publishable_Work_Summary.tex §moments.

From $I' = -\delta J$ and the Riccati:

Result 7.1 (First and second moments; mean release).

$$J(t) = \mathbb{E}[X_t] = \frac{(a-b)^2 w}{(B+Aw)^2} = -\frac{\beta}{\delta}(I-a)(I-b), \quad (11)$$

$$K(t) = \mathbb{E}[X_t^2] = \left[1 + \frac{2\beta}{\delta}(1-I)\right]J = \frac{2\beta}{\delta}(\kappa-I)J, \quad \kappa = 1 + \frac{\delta}{2\beta}, \quad (12)$$

$$V(t) = \mathbb{E}[W_t] = (1-I)\left[1 + \frac{\beta}{\delta}(1-I)\right]. \quad (13)$$

Remark 7.2 (Simplifications). • Thesis J carries a cluttered prefactor; $AB = \delta/\beta$ cancels it and $J(0) = 1$ is manifest.

- Thesis $V = \frac{\beta-\mu}{\delta}(1-I) - J + 1$ equals (13), a **quadratic in $(1-I)$ alone** — no separate J or μ .
- Consequently $V' = \delta K$, so $g := \delta K$ is the release flux.
- Alternative: $V = \frac{\beta}{\delta}[(I-\kappa)^2 - (1-\kappa)^2]$.

Result 7.3 (Lifetime-time release and conditional mean burst size).

$$V_\infty = \frac{a(1-b)}{a-1} = \frac{aB}{A} = \frac{\beta-\mu}{\delta}(1-b) + 1, \quad \mathbb{E}[\mathcal{K} \mid \text{burst}] = \frac{V_\infty}{1-b} = \frac{a}{a-1}. \quad (14)$$

V_∞ includes *realisations that clear internally and release nothing (mass b at 0)*. The conditional mean removes them. When $\mu = 0$: $b = 0$, $a = 1 + \delta/\beta$, $V_\infty = 1 + \beta/\delta = \langle X \rangle_{\text{QS}}$. When $\mu > 0$ they differ (e.g. (1, 0.2, 0.05): 17.23 vs 13.99).

8 Second moment of release [CORRECTED]

[SOURCE] raw (wrong signs): 3 BDC core/document MAIN/sections/08_variance_and_standard_deviation_of.tex corrected formula: new_notes/BMVR_extension_notes.tex Part I §“Second moment of the release”.

Starting from $d\mathbb{E}[W_t^2]/dt = \delta\mathbb{E}[X_t^3]$ and eliminating the third moment via the K ODE, with correct initial data $(I, J, K, V, \mathbb{E}[W^2])(0) = (1, 1, 1, 0, 0)$:

Result 8.1 (Second moment of cumulative release).

$$\mathbb{E}[W_t^2] = \frac{2(\beta - \mu)}{\delta} V - \frac{\beta + \mu}{\delta} I - K + \frac{\beta + \mu}{\delta} + 1. \quad (15)$$

$$\text{Var}(W_t) = \mathbb{E}[W_t^2] - V^2.$$

Remark 8.2 ([**CORRECTED**] thesis errors). Thesis had wrong signs on I and K , and omitted $+1$ (from $K(0) = 1$, not 0). Cause: prose used $J = +\delta^{-1}I'$ instead of $J = -\delta^{-1}I'$. Independent check $\mu = 0$, $t \rightarrow \infty$: geometric on $\{1, 2, \dots\}$ with success $s = \delta/(\beta + \delta)$ gives $\mathbb{E}[W^2] = (2 - s)/s^2 = 2\beta^2/\delta^2 + 3\beta/\delta + 1$, matching (15).

Remark 8.3 ([**CORRECTED**] K as function of I). Correct form (Ch.3 §11, not §9): $K = \frac{2\beta}{\delta}(\kappa - I)J$ with $J = -\frac{\beta}{\delta}(I - a)(I - b)$. The §9 version has wrong power of δ and a sign error. Prefer the (I, J) form (12) over expanded polynomials.

Part B — Burst time, size, and the QSD

9 Burst time

[**SOURCE**] implicit in 3 BDC core/document MAIN/sections/05_analytical_quantities.tex ($I' = -\delta J$); named in new_notes/BMVR_extension_notes.tex, new_notes2/Core_Publishable_Work_Summary.tex; alluded to but not written in document MAIN/sections/09_potential_application_in_bmvr.tex.

Result 9.1. $\mathbb{P}\{\tau > t\} = I(t)$, so the (defective) density of burst time is

$$\varphi(t) = -I'(t) = \delta J(t), \quad \int_0^\infty \varphi(t) dt = 1 - b.$$

Mass b sits at $\tau = \infty$ (internal clearance without burst).

10 State probabilities are geometric at every time

[**SOURCE**] document MAIN/sections/05_state_probabilities.tex; QSD discussion: document MAIN/sections/06_quasi_stationary_distribution.tex.

For $n \geq 1$,

$$p_n(t) = \left(\frac{a-b}{b-aw}\right)^2 w P(t)^{n-1}, \quad P(t) = \frac{1-w}{b-aw}, \quad w = e^{\theta t}. \quad (16)$$

Conditional on $X_t \geq 1$, the load is geometric with ratio $P(t)$ at every t , and $P(t) \rightarrow 1/a$.

11 Burst size equals the quasi-stationary distribution

[**NEW**] ingredients in document MAIN/sections/05_state_probabilities.tex, 06_quasi_stationary_distribution.tex, 07_burst_size_distribution.tex; identity stated in new_notes/BMVR_extension_notes.tex, new_notes2/Core_Publishable_Work_Summary.tex.

Result 11.1 (Burst size = QSD). Both are geometric on $\{1, 2, \dots\}$ with ratio $1/a$:

$$\text{QSD}(n) = \lim_{t \rightarrow \infty} \mathbb{P}\{X_t = n \mid X_t \notin \{0, R\}\} = (a-1) a^{-n}, \quad (17)$$

$$\mathbb{P}\{\mathcal{K} = k \mid \text{burst}\} = (a - 1) a^{-k}. \quad (18)$$

Unconditionally (including no-burst realisations), $\mathbb{P}\{\mathcal{K} = k\} = (\delta/\beta) a^{-k}$ for $k \geq 1$, with mass b at 0.

Result 11.2 (Proof of the burst-size law — no simulation required). *With $P(t) = (1-w)/(b-aw)$,*

$$\frac{d}{dt} \left[\frac{P^k}{k\beta} \right] = p_k(t).$$

Hence

$$\mathbb{P}\{\mathcal{K} = k\} = \delta k \int_0^\infty p_k(t) dt = \frac{\delta}{\beta} [P^k]_0^\infty = \frac{\delta}{\beta} a^{-k}.$$

Checks: (i) $\sum_{k \geq 1} (\delta/\beta) a^{-k} = 1 - b$; (ii) $\sum_{k \geq 1} k(\delta/\beta) a^{-k} = V_\infty$.

Immediate QSD moments:

$$\langle X \rangle_{\text{QS}} = \frac{a}{a-1}, \quad \text{Var}_{\text{QS}}(X) = \frac{a}{(a-1)^2}, \quad \langle X^2 \rangle_{\text{QS}} = \frac{a(a+1)}{(a-1)^2}. \quad (19)$$

Remark 11.3 ([**OPEN**] conceptual proof). Algebra is complete. A structural argument (size-biasing composed with burst intensity, or Yaglom-type limit) would make the identity quotable rather than coincidental.

12 Size-biasing and late bursts

A cell bursts at rate δX_t , so given burst at time t the size is size-biased:

$$\mathbb{P}\{\mathcal{K} = k \mid \tau = t\} = \frac{k p_k(t)}{J(t)}, \quad \mathbb{E}[\mathcal{K} \mid \tau = t] = \frac{K(t)}{J(t)} \xrightarrow[t \rightarrow \infty]{} 1 + \frac{2\beta}{\delta}(1-b).$$

A very late burst is roughly twice the size of an average one.

13 Resolving Ch.4 §8 “curious circularity”

[**CORRECTED**] [SOURCE] document MAIN/sections/08_further_considerations.tex; resolution in new_notes/BMVR_extension_notes.tex Part I.

Ch.4 computes $\mathbb{E}[\mathcal{K} \mid \tau > t] = (\mathbb{E}[\mathcal{K}] - V(t))/I(t)$, finds the $t \rightarrow \infty$ limit returns $\mathbb{E}[\mathcal{K}]$, and records confusion.

With consistent definitions:

$$\mathbb{E}[\mathcal{K} \mathbf{1}\{\text{burst}\} \mid \tau > t] = \frac{V_\infty - V(t)}{I(t)} \rightarrow 0, \quad (20)$$

correct: conditioning on “has not burst yet” converges to “never bursts”.

The interesting limit conditions on eventual burst:

$$\mathbb{E}[\mathcal{K} \mid \tau > t, \text{ burst eventually}] = \frac{V_\infty - V(t)}{I(t) - b} \rightarrow \frac{\langle X^2 \rangle_{\text{QS}}}{\langle X \rangle_{\text{QS}}}, \quad (21)$$

by l'Hôpital ($V' = \delta K$, $I' = -\delta J$).

14 Mean productive lifetime

Result 14.1.

$$\mathbb{E}[T_{\text{prod}}] = \int_0^\infty \hat{I}(t) dt = \frac{1}{\beta} \log\left(\frac{a}{a-1}\right) = \frac{1}{\beta} \log\langle X \rangle_{\text{QS}}. \quad (22)$$

When $\mu = 0$: $\beta^{-1} \log(1 + \beta/\delta)$. Interpretation: time for exponential growth at rate β to reach the quasi-stationary level.

Part C — Multiplicity of infection and chained transfer

15 MOI kernels [CORRECTED]

[SOURCE] $\hat{I}_k$ wrong form flagged in 3 BDC core/document MAIN/sections/05_analytical_quantities.tex corrected kernels and $V_\infty^{(k)}$ developed in this document (and partially in new_notes2/Core_Publishable_Work_Sheet note MOI g_k there lacked I -powers — use formulas here).

For one cell founded by k particles (shared catastrophe clock):

Result 15.1 (Correct MOI moments and release kernel).

$$\hat{I}_k(a) = I(a)^k - D(a)^k, \quad (23)$$

$$J_k(a) = k I(a)^{k-1} J(a), \quad (24)$$

$$K_k(a) = k K(a) I(a)^{k-1} + k(k-1) J(a)^2 I(a)^{k-2}, \quad (25)$$

$$g_k(a) = \delta K_k(a). \quad (26)$$

Remark 15.2 ([CORRECTED] what was wrong). Earlier notes wrote $g_k = \delta[kK + k(k-1)J^2]$ (free sum of independent counts). That omits the I -powers from the shared survival indicator. Master-equation check at $(1, 0.2, 0.05)$, $t = 1$, $k = 2$: wrong $K_2 \approx 23.39$; correct $K_2 = 22.264$ matches ME exactly.

Result 15.3 (Lifetime mean release from k founders).

$$V_\infty^{(1)} = V_\infty = \frac{a(1-b)}{a-1}, \quad (27)$$

$$V_\infty^{(k)} = (1-b^k) + \frac{2\beta}{\delta} \left[\frac{1}{k+1} - b^k + \frac{k b^{k+1}}{k+1} \right] + k(k-1) \frac{\beta}{\delta} \int_1^b (x-a)(x-b) x^{k-2} dx \quad (k \geq 2). \quad (28)$$

Special case $\mu = 0$:

$$V_\infty^{(k)} = k + \frac{\beta}{\delta} \quad (29)$$

(e.g. $\beta = 1, \delta = 0.1$: 11, 12, 13, 14, 15, ...).

Lifetime yield is **not** kV_∞ (shared first-catastrophe dumps the whole cell once). Conditional mean given eventual burst increases with k .

[NEW] Superlinearity of g_k in k is a testable single-cell / low-MOI prediction; classical BMVR cannot express it.

16 Chained immediate-transfer model ($N = 2$ and general M)

[SOURCE] N=2 Immediate Transfer content/N_eq_2.tex (PDF: N=2 Immediate Transfer content/N_eq_2.pdf).

Setup. M host cells. One starts with a single particle. On rupture, the entire load is transferred immediately into a new empty cell (no extracellular phase). Process continues until M ruptures. Extreme caricature of serial infection; useful for tractable multi-cell burst statistics.

Notation. r_i = load at the i th rupture; t_i = time of i th rupture; $T_i = t_i - t_{i-1}$ = inter-rupture interval. Birth rate λ in source text = our β .

Rupture-size laws ($\mu = 0$). With $s = \delta/(\delta + \beta)$ and $\rho = \beta/(\delta + \beta)$,

$$\mathbb{P}\{r_k = n\} = \mathcal{A}_k^{(n)} s^k \rho^{n-1}, \quad \mathcal{A}_k^{(n)} = \frac{(n+k-2)!}{(k-1)!(n-1)!} = \binom{n+k-2}{k-1}. \quad (30)$$

Special cases:

$$\begin{aligned} \mathbb{P}\{r_1 = n\} &= s \rho^{n-1} && \text{(geometric — matches one-cell BDC, } \mu = 0\text{),} \\ \mathbb{P}\{r_2 = n\} &= n s^2 \rho^{n-1}, \\ \mathbb{P}\{r_3 = n\} &= \frac{n(n+1)}{2} s^3 \rho^{n-1}. \end{aligned}$$

Recurrence: $\mathcal{A}_k^{(n)} = \sum_{i=1}^n \mathcal{A}_{k-1}^{(i)}$ with $\mathcal{A}_1^{(i)} = 1$.

First rupture time from m founders. If $S(t) = I(t)$ is the no-burst probability from one founder, then

$$\mathbb{P}\{t_1 > t\} = S(t)^m, \quad \text{density } \mathcal{F}_m(t) = -m S(t)^{m-1} S'(t).$$

[OPEN] Full joint laws of (t_2, r_2) and higher; interval distributions for general μ ; link of chained r_k laws to MOI kernel g_k under immediate transfer.

Part D — From one cell to a population: renewal BMVR

17 Why constant- p substitution fails

[SOURCE] document MAIN/sections/09_potential_application_in_bmvr.tex; diagnosis and replacement in new_notes/BMVR_extension_notes.tex Part II, new_notes2/Core_Publishable_Work_Su §“From one cell to a population”.

Ch.4 proposed replacing BMVR release rate p by $\langle X \rangle_{\text{QS}}$.

Obstruction is dimensional. p has units particles/time; $\langle X \rangle_{\text{QS}}$ is a pure count. When $\mu = 0$, $\langle X \rangle_{\text{QS}} = V_\infty$, so the proposal substitutes total lifetime output for a rate.

Candidate integrals also fail:

- $\int \varphi(t) \mathbb{E}[\mathcal{K} \mid \tau = t] dt = V_\infty$ — a count, not a rate.
- Weighting by the burst-time density is wrong; the correct weight is the **age distribution of currently infected cells**.

Further: $\mathcal{V}(t) = N_0 V(t)$ only for synchronously infected cells. Asynchrony is the whole problem — and the whole solution (renewal).

18 Classical BMVR (comparator)

[SOURCE] standard literature (Nowak–Bangham; Perelson et al.); written for comparison in `new_notes2/Core_Publishable_Work_Summary.tex` §“Classical BMVR”; erroneous infection term $\propto I$ flagged in `new_notes/BMVR_extension_notes.tex` Part IV (Ch.1).

Simplified (target-cell constant) Basic Model of Viral Replication:

$$\begin{aligned}\frac{d\mathcal{I}}{dt} &= \gamma T \mathcal{V} - d_{\mathcal{I}} \mathcal{I}, \\ \frac{d\mathcal{V}}{dt} &= p \mathcal{I} - c \mathcal{V}.\end{aligned}\tag{31}$$

Both p and $d_{\mathcal{I}}$ are phenomenological.

[CAVEAT] Ch.1 thesis draft sometimes wrote infection as βSI (driven by infected cells). Correct driving variable is free virus: $\gamma T \mathcal{V}$.

19 The two BDC kernels

[SOURCE] `new_notes/BMVR_extension_notes.tex` Part II §kernels; `new_notes2/Core_Publishable_Work_Summary.tex` §“The two kernels from the BDC”; closed forms from Part A of this document.

$$\hat{I}(a) = \frac{(a-b)^2 w}{(B+Aw)(aw-b)}, \quad w = e^{\theta a},\tag{32}$$

$$g(a) = \delta K(a) = V'(a) = \text{expected release flux at cell-age } a.$$

Release kernel: burst rate δX_t times size X_t gives expected flux $\delta \mathbb{E}[X_t^2] = \delta K$.

20 Burst-aware (renewal) viral dynamics

[SOURCE] `new_notes/BMVR_extension_notes.tex` Part II §“The correct object”; `new_notes2/Core_Publishable_Work_Summary.tex` (Result “New viral dynamics — burst-aware BMVR”). Numerical verification: `new_notes3/verify_result_20_1.tex` (Section 21).

Result 20.1 (New viral dynamics — burst-aware BMVR). *With incidence $i(t) = \gamma T \mathcal{V}(t)$,*

$$\mathcal{I}(t) = \mathcal{I}_0 \hat{I}(t) + \int_0^t i(t-\alpha) \hat{I}(\alpha) d\alpha = \mathcal{I}_0 \hat{I} + (i * \hat{I}),\tag{33}$$

$$\frac{d\mathcal{V}}{dt} = \mathcal{I}_0 g(t) + \int_0^t i(t-\alpha) g(\alpha) d\alpha - c \mathcal{V}(t) = \mathcal{I}_0 g + (i * g) - c \mathcal{V}.\tag{34}$$

What each equation replaces.

Classical (31)	New (33)–(34)
$\mathcal{I}' = i - d_{\mathcal{I}}\mathcal{I}$ (constant removal)	$\mathcal{I} = \mathcal{I}_0\hat{I} + i * \hat{I}$ (age-structured survival)
$\mathcal{V}' = p\mathcal{I} - c\mathcal{V}$ (constant release)	$\mathcal{V}' = \mathcal{I}_0g + i * g - c\mathcal{V}$ (age-structured release)

Remark 20.2 (Novelty claim — state honestly). The renewal/age-structured framework is standard (McKendrick–von Foerster; Nelson–Gilchrist–Perelson-type models). The novelty is that $\hat{I}$ and g are **derived in closed form from a mechanistic intracellular CTMC**, plus the consequences below (p_{eff} , flooding, etc.).

Optional differential form for $\mathcal{I}$.

$$\mathcal{I}' = i + \mathcal{I}_0\hat{I}' + i * \hat{I}'.$$

No constant $d_{\mathcal{I}}$ reproduces this for all histories of i ; reduction to classical form holds only for pure exponentials (next section).

21 Numerical verification of Result 20.1

A dedicated suite (this folder) `new_notes3/verify_result_20_1.py` checks Result 20.1 end-to-end. Full machine log: `new_notes3/verify_result_20_1_report.txt`; figures in `new_notes3/verify_figure`. **Score: 53/53 checks passed** (wall time ~ 17 s on a laptop).

21.1 What is being verified

Result 20.1 is a linear age-structured mean-field system. The suite tests four layers:

1. **Kernel integrity** — inputs $\hat{I}, g = \delta K$ obey single-cell identities.
2. **IDE self-consistency** — numerical solutions satisfy the integral equations to discretisation error.
3. **Structural limits** — exponential kernels $\Rightarrow$ classical BMVR; $\gamma T = 0 \Rightarrow$ pure single-cell convolution; R_0 threshold and characteristic growth rate.
4. **Stochastic ground truth (cohort)** — Gillespie ensemble of independent BDC cells with free-virus clearance, $\gamma T = 0$, matches the renewal means.

[CAVEAT] With reinfection ($\gamma T > 0$) the renewal system is the *exact mean-field* for independent cells and linear incidence; a full stochastic multi-generation Gillespie check of means is left as a heavier extension. Tests E–G already constrain the reinfection case through the characteristic equation, which any correct linear theory must satisfy.

21.2 Test catalogue and outcomes

ID	Name	What is checked	Result
A	Kernel self-consistency	$V' = g, \int g = V_\infty, \int \hat{I} = \mathbb{E}[T_{\text{prod}}], I' = -\delta J$ (5 param. sets)	5/5 PASS
B	$\gamma T = 0$ cohort	$\mathcal{I} = \mathcal{I}_0 \hat{I}; \mathcal{V}' = \mathcal{I}_0 g - c\mathcal{V}$ vs exact convolution; $c = 0 \Rightarrow \mathcal{V} = \mathcal{I}_0 V(t)$	15/15 PASS
C	V -equation residual	$\mathcal{V}' - \mathcal{I}_0 g - \gamma T(\mathcal{V} * g) + c\mathcal{V} \approx 0$ on supercritical / subcritical / V_0 -seed solutions	5/5 PASS
D	Exp. kernels $\equiv$ classical	$S = e^{-da}, g = pe^{-da} \Rightarrow$ renewal matches BMVR ODE (rel. err. $< 5 \times 10^{-5}$)	3/3 PASS
E	Growth = char. root	late log $\mathcal{V}$ slope = r_* solving $r + c = \gamma T \tilde{g}(r)$ (rel. err. $\sim 10^{-6}$ – 10^{-7})	5/5 PASS
F	R_0 threshold	$R_0 \in \{0.5, \dots, 3\}$: decay iff $R_0 < 1$, growth iff $R_0 > 1$	8/8 PASS
G	$p_{\text{eff}}(r_*)$ match	classical ODE with $(p, d) = (p_{\text{eff}}(r_*), d_{\text{eff}}(r_*))$ shares r_* with renewal	4/4 PASS
H	Gillespie cohort	independent BDC cells + clearance; ensemble mean vs renewal (rel. $L^2 \sim 1$ – 2%)	2/2 PASS
I	V_0 -only seed	early $\mathcal{V} \approx V_0 e^{-ct}; \mathcal{I}(0) = 0$ then rises	1/1 PASS
J	Lifetime output	$\gamma T = c = 0: \mathcal{V}(\infty) = V_\infty$ (rel. err. $< 10^{-6}$)	5/5 PASS
K	Well-posedness	non-negative finite trajectories under matched comparison	1/1 PASS
Total			53/53 PASS

21.3 Headline quantitative findings

A — Kernels. Across $(\beta, \mu, \delta) \in \{(1, 0, 0.1), (1, 0.2, 0.05), (1, 0, 0.5), (1, 0.9, 0.1), (1, 0.5, 1/3)\}$: $\max |V' - g| / \max g \leq 2 \times 10^{-4}$, $|\int g - V_\infty| / V_\infty \leq 10^{-8}$, $|\int \hat{I} - \mathbb{E}[T_{\text{prod}}]| / \mathbb{E}[T_{\text{prod}}] \leq 2 \times 10^{-8}$.

B — No reinfection. $\mathcal{I}$ matches $\mathcal{I}_0 \hat{I}$ to machine precision (0). $\mathcal{V}$ matches the forced linear ODE / convolution to $\leq 4 \times 10^{-6}$ relative; with $c = 0$, $\mathcal{V} = \mathcal{I}_0 V_{\text{single}}(t)$ to $\sim 10^{-6}$.

C — Residuals with reinfection. On five scenarios (including subcritical and free-virion seed), max relative V -equation residual $\leq 3.4 \times 10^{-4}$; $\mathcal{I}$ definition residual $\leq 10^{-16}$ (by construction of the solver).

D — Exponential reduction (structural correctness).

(p, d)	$\max \mathcal{V}^{\text{ren}} - \mathcal{V}^{\text{cls}} / \max$	same for $\mathcal{I}$
(2.0, 0.5)	1.8×10^{-5}	1.8×10^{-5}
(4.5, 0.4)	4.7×10^{-5}	4.8×10^{-5}
(1.0, 1.0)	1.3×10^{-5}	9.9×10^{-6}

When kernels are pure exponentials, Result 20.1 is classical BMVR. Figure: `new_notes3/verify_figures/D_exp`

E — Characteristic growth rate.

Case	R_0	r_* (char. eq.)	r_{fit} (from $\mathcal{V}$)	rel. err.
A ($\mu = 0$)	1.760	0.08775	0.08775	3×10^{-7}
B ($\mu > 0$)	2.098	0.09040	0.09040	2×10^{-7}
C (fast δ)	2.500	0.24446	0.24446	1×10^{-6}
E (high μ)	2.632	0.17021	0.17021	7×10^{-7}
F (V_0 seed)	2.750	0.15132	0.15132	5×10^{-7}

Figure: `new_notes3/verify_figures/E_growth_rate_match.pdf`.

F — Threshold. For fixed $(\beta, \mu, \delta, c) = (1, 0, 0.1, 0.25)$ and R_0 scanned through 1: subcritical trajectories decay ($r_{\text{fit}} < 0$); supercritical grow with $r_{\text{fit}} = r_* > 0$. At $R_0 = 1.05$, $r_* = 0.00694$ already resolved. Figure: `new_notes3/verify_figures/F_R0_threshold.pdf`.

G — Effective-parameter projection. Classical BMVR with $p = p_{\text{eff}}(r_*)$, $d = d_{\text{eff}}(r_*)$ reproduces the renewal growth rate to relative error $\sim 10^{-6}$ – 10^{-7} (as theory requires). Transients still differ — only the exponential phase is matched.

H — Gillespie vs mean-field ($\gamma T = 0$).

	$n_{\text{reps}} \times \mathcal{I}_0$	rel. $L^2(\mathcal{I})$	rel. $L^2(\mathcal{V})$
$\mu = 0, \delta = 0.1, c = 0.25$	400×20	0.89%	1.81%
$\mu = 0.2, \delta = 0.05, c = 0.2$	500×30	0.61%	1.34%

Ensemble means of productive infected count and free virus (impulse releases at burst times + exponential clearance) track the renewal solution within sampling error. Figures: `new_notes3/verify_figures/H_gillespie_mu_pos.pdf`.

I–K — Seeds, lifetime, positivity. Early free-virion seed: $\max |\mathcal{V} - V_0 e^{-ct}|/V_0 \approx 10^{-3}$ on $[0, 0.5]$. Lifetime output $\mathcal{V}(\infty) = V_\infty$ to $< 10^{-6}$ relative. Solutions remain non-negative and finite.

21.4 Verdict on Result 20.1

Result 20.1 is numerically verified at the level appropriate to a linear renewal mean-field model:

- kernels are consistent with single-cell theory;
- the IDE is solved accurately and satisfies its own residual tests;
- every structural limit that can be checked in closed form holds (exponential kernels, $\gamma T = 0$, R_0 threshold, $r = r_*$, p_{eff});
- independent stochastic simulation of a finite cohort reproduces the mean-field curves for $\mathcal{I}$ and $\mathcal{V}$ when there is no reinfection.

Remaining (optional) extension: multi-generation Gillespie with reinfection to confirm mean trajectories for $\gamma T > 0$ at large population size — expected to agree by linearity, but not yet coded.

21.5 How to reproduce

```
cd "4 BDC additional and BMVR/new_notes3"
python3 verify_result_20_1.py
# writes verify_result_20_1_report.txt, metrics.json, verify_figures/*.pdf
```

22 Numerical overlays (already generated)

[SOURCE] script `figures/plot_bmvr_comparison.py`; outputs `figures/kernels.pdf`, `overlay_*.pdf`; described in `new_notes2/Core_Publishable_Work_Summary.tex` §“Numerical overlays of the two systems”.

Script `figures/plot_bmvr_comparison.py`. Reading code for all plots:

- **Solid navy** = NEW (renewal / burst-aware).
- **Dashed orange with circles** = CLASSICAL matched at $r = 0$:

$$p = p_{\text{eff}}(0) = \frac{\beta V_{\infty}}{\log(a/(a-1))}, \quad d_{\mathcal{I}} = \frac{\beta}{\log(a/(a-1))}.$$

- Dotted green (optional) = classical with naive $p = \langle X \rangle_{\text{QS}}$ (dimensionally wrong; systematically wrong scale).

Even with fair matching, classical ODEs miss early lag, peak timing, and late growth under strong age-structure. Relative free-virion discrepancy is often order-one in supercritical regimes.

23 Effective parameters, R_0 , phase-dependent p

[SOURCE] `new_notes/BMVR_extension_notes.tex` Part II §“Effective BMVR parameters”, §“Characteristic equation, R_0 ”; `new_notes2/Core_Publishable_Work_Summary.tex` §“Effective parameters, R_0 , and growth-phase dependence”.

Result 23.1 (Effective BMVR parameters). *Writing $\tilde{f}(r) = \int_0^{\infty} e^{-r\alpha} f(\alpha) d\alpha$:*

$$p_{\text{eff}}(r) = \frac{\delta \tilde{K}(r)}{\tilde{I}(r)}, \quad d_{\mathcal{I},\text{eff}}(r) = \frac{1}{\tilde{I}(r)} - r. \quad (35)$$

Exact for exponential solutions $i \propto e^{rt}$; not for full transients. Substituting into the classical eigenvalue relation recovers the renewal characteristic equation.

Limits:

$$r = 0: \quad p_{\text{eff}} = \frac{\beta V_{\infty}}{\log(a/(a-1))}, \quad d_{\mathcal{I},\text{eff}} = \frac{\beta}{\log(a/(a-1))}; \quad (36)$$

$$r \rightarrow \infty: \quad p_{\text{eff}} \rightarrow \delta, \quad d_{\mathcal{I},\text{eff}} \rightarrow \mu + \delta \quad (\text{young-cell: one particle, burst rate } \delta). \quad (37)$$

$p_{\text{eff}}(r)$ is decreasing in r (checked across representative parameters). At $(1, 0, 0.1)$: $p_{\text{eff}}(0) \approx 4.59$ vs $p_{\text{eff}}(\infty) = 0.1$.

Closed Laplace transforms [NEW] (hypergeometric). With $s := r/\theta$, $z := -B/A$:

$$\tilde{I}(r) = \frac{1}{\beta(s+1)} \left[\frac{B}{A} {}_2F_1\left(1, s+1; s+2; -\frac{B}{A}\right) + \frac{b}{a} {}_2F_1\left(1, s+1; s+2; \frac{b}{a}\right) \right], \quad (38)$$

$$\delta\tilde{K}(r) = \frac{\delta}{\beta} \left[\frac{{}_2F_1(1, s; s+2; z)}{A(s+1)} + \frac{{}_2F_1(2, s; s+3; z)}{A^2(s+1)(s+2)} \right]. \quad (39)$$

At $s = 0$ both recover $\int \hat{I} = \beta^{-1} \log(a/(a-1))$ and $\delta\tilde{K}(0) = V_\infty$.

Result 23.2 (R_0 unchanged by bursting at fixed total output). *Characteristic equation:*

$$r + c = \gamma T \delta\tilde{K}(r), \quad \text{hence} \quad R_0^{\text{ODE}} = \frac{\gamma T V_\infty}{c}.$$

At fixed V_∞ , bursting changes generation times (hence r and extinction) but not this threshold.

Result 23.3 (Prediction for fitting practice). *BMVR fits to acute-phase data and set-point data should yield systematically different p , in a direction and amount the theory specifies. The eclipse-like delay usually inserted by hand emerges here from intracellular dynamics with no extra parameter.*

Result 23.4 (Identifiability — negative result). *Three rates (β, μ, δ) map to two BMVR parameters (p, d_I) at any fixed r . Not invertible. Third observables that over-determine the map:*

1. burst-size dispersion (geometric ratio $1/a$);
2. fraction of cells that clear without bursting (b);
3. shape of release-time density $\varphi = \delta J$.

Ch.4 §9's aim — recover intracellular rates from p alone — is impossible.

[CAVEAT][Two R_0 notions]

- ODE / renewal continuum: $R_0^{\text{ODE}} = \gamma T V_\infty / c$ (one virion can contribute mean $\gamma T / c$ infections).
- Discrete branching with virion absorption: $m = q V_\infty$, $q = \gamma T / (\gamma T + c)$.

They agree when $\gamma T \ll c$. Flooding calculations use the branching mean $m = q V_\infty$; write “ R_0 ” carefully in each context.

Part E — Flooding advantage and growth-rate trade-off

24 Offspring of one infected cell

[SOURCE] `new_notes/BMVR_extension_notes.tex` Part II §“Offspring distribution of one infected cell”; `new_notes2/Core_Publishable_Work_Summary.tex` §“Offspring of one infected cell”.

Each released particle infects with probability $q = \gamma T / (\gamma T + c)$. With $y = 1 - q + qz$:

$$G_{\text{off}}(z) = b + \frac{\delta}{\beta} \frac{y}{a - y}, \quad G_{\text{off}}(1) = 1, \quad m = G'_{\text{off}}(1) = qV_{\infty}. \quad (40)$$

Variance:

$$\text{Var}_{\text{burst}} = \frac{2q^2 V_{\infty}}{a - 1} + qV_{\infty} - q^2 V_{\infty}^2.$$

25 Extinction probability in closed form

$G_{\text{off}}(z) = z$ is quadratic in y ; the non-trivial root is

$$z_{\text{ext}}^{\text{burst}} = \frac{a(1 - q + qb) - 1 + q}{q} = \frac{a - 1}{q} + 1 - a(1 - b). \quad (41)$$

26 Flooding advantage — main theorem

[SOURCE] `new_notes/BMVR_extension_notes.tex` Part II §“The flooding advantage, answered exactly”; `new_notes2/Core_Publishable_Work_Summary.tex` §“The flooding advantage”; $L = 1$ law-match and r -trade-off strengthened in this document.

Comparator: BMVR cell with $\text{Exp}(d_{\mathcal{I}})$ lifetime, constant rate p , matched so $p/d_{\mathcal{I}} = V_{\infty}$ and same q . Offspring is geometric with mean $m = qV_{\infty}$, so $z_{\text{ext}}^{\text{bud}} = 1/m$ when $m > 1$.

Write $L := a(1 - b) = a - \mu/\beta$, so $V_{\infty} = L/(a - 1)$.

Result 26.1 (Flooding advantage). *For $m > 1$,*

$$z_{\text{ext}}^{\text{burst}} - z_{\text{ext}}^{\text{bud}} = (L - 1) \left(\frac{1}{m} - 1 \right). \quad (42)$$

Hence bursting strictly lowers extinction probability relative to matched-mean budding iff $L > 1$, i.e.

$$\boxed{\frac{1}{\delta} < \frac{1}{\beta} + \frac{1}{\mu}} \quad (\text{when } \mu = 0: \text{ always, for all } \delta > 0). \quad (43)$$

When $L = 1$ the extinction probabilities are identical for every q .

Result 26.2 ([NEW] strengthening at $L = 1$). *When $L = 1$ (equivalently $b = \delta/\beta$), the offspring law is **exactly geometric** with mean m — not merely same extinction probability. Hence variances match as well. Equivalence chain:*

$$L = 1 \iff a = 1 + \frac{\mu}{\beta} \iff \delta(\beta + \mu) = \beta\mu \iff \frac{1}{\delta} = \frac{1}{\beta} + \frac{1}{\mu}.$$

Result 26.3 ([NEW] variance ordering). *At matched mean m , $\text{Var}_{\text{burst}} < \text{Var}_{\text{bud}}$ iff $L > 1$, with equality at $L = 1$ (geometric case), and the reverse when $L < 1$.*

Numerical checks (exact).

(β, μ, δ)	q	z^{burst}	z^{bud}	Winner
(1, 0, 0.1)	0.2	0.400	0.455	bursting ($L > 1$)
(1, 0.9, 0.1)	0.9	0.935	0.844	budding ($L < 1$)
(1, 0.5, 1/3)	0.6	0.833	0.833	tie ($L = 1$, laws equal)

Interpretation. Bursting concentrates output (lower offspring variance when $L > 1$) but carries all-or-nothing risk of internal clearance (probability b). Advantage survives when the burst timescale $1/\delta$ is shorter than the harmonic-type combination of birth and death timescales.

[CAVEAT] Comparison holds the offspring *mean* fixed — branching structure only. Nonlinear effects (immune saturation, local depletion, Allee) are separate; name both mechanisms in any write-up (Komarova-type discussion).

27 Deterministic growth-rate trade-off [\[NEW\]](#)

[SOURCE] numerical comparison developed for this document (not in Ch.3/4; related discussion of R_0 vs r in `new_notes/BMVR_extension_notes.tex` Part II).

At matched R_0^{ODE} (or matched m) **and** matched mean productive lifetime $d = 1/\mathbb{E}[T_{\text{prod}}]$, $p = V_\infty d$:

Result 27.1 (Budding is faster in the exponential phase). *In all regimes checked* ($L > 1$, $L = 1$, $L < 1$),

$$r_{\text{bud}} > r_{\text{burst}}.$$

Example: $(\beta, \mu, \delta) = (1, 0, 0.1)$, $R_0 = 2$: $r_{\text{burst}} \approx 0.18$, $r_{\text{bud}} \approx 0.25$.

Biological headline for a paper: at fixed total output per cell, bursting can *lower early extinction* (when $L > 1$) while *slowing deterministic invasion speed* (generation-time delay). R_0 fixed; z_{ext} and r can move in opposite directions.

28 Optimal burst rate at fixed mean offspring [\[NEW\]](#)

At fixed $m = R_0$ (adjust $q = m/V_\infty$) and $\mu = 0$: faster bursting (larger δ) lowers z_{ext} until V_∞ becomes too small to support the target m with $q \leq 1$.

At fixed m and $\mu > 0$: bursting beats budding only once δ crosses the $L = 1$ threshold $\delta_* = \beta\mu/(\beta + \mu)$; below that threshold, budding wins on extinction.

Part F — Spectrum of BMVR equation variations

29 Universal renewal skeleton

Any intracellular model that supplies a survival kernel $S(a)$ and release kernel $g(a)$ induces

$$i(t) = \gamma T \mathcal{V}(t) \quad (\text{or a nonlinear incidence } f(\mathcal{V}, T)), \quad (44)$$

$$\mathcal{I}(t) = \mathcal{I}_0 S(t) + (i * S)(t), \quad (45)$$

$$\mathcal{V}'(t) = \mathcal{I}_0 g(t) + (i * g)(t) - c \mathcal{V}(t). \quad (46)$$

Effective parameters: $p_{\text{eff}}(r) = \tilde{g}(r)/\tilde{S}(r)$, $R_0 = (\gamma T/c) \int_0^\infty g$ (linear incidence).

Model	$S(a)$	$g(a)$
Classical BMVR	e^{-da}	$p e^{-da}$
Absorbing BDC	$\hat{I}(a)$	$\delta K(a)$
MOI- k BDC	$I^k - D^k$	δK_k (corrected)
Reset catastrophe	e^{-da}	$\delta e^{-da} \mathbb{E}[X_a^2]$
HIV multi-stage	stage survival	$p \times$ productive occupancy
Two-type / GATE	two-type $\hat{I}$	two-type second-moment flux

30 Five modular ODE cases (mean-field ladder)

[SOURCE] new_notes2/Core_Publishable_Work_Summary.tex §“Building BMVR piece by piece: five cases”.

Case 1 — Classic BMVR.

$$\mathcal{I}' = \gamma T \mathcal{V} - d_{\mathcal{I}} \mathcal{I}, \quad \mathcal{V}' = p \mathcal{I} - c \mathcal{V}. \quad R_0 = \frac{\gamma T p}{c d_{\mathcal{I}}}.$$

Case 2 — Eclipse + classic release.

$$E' = \gamma T \mathcal{V} - (a + d_E) E, \quad \mathcal{I}' = a E - d_{\mathcal{I}} \mathcal{I}, \quad \mathcal{V}' = p \mathcal{I} - c \mathcal{V}.$$

$R_0 = \frac{\gamma T}{c} \cdot \frac{a}{a+d_E} \cdot \frac{p}{d_{\mathcal{I}}}$. Multi-stage (Erlang) eclipse: chain $E_1 \rightarrow \dots \rightarrow E_n \rightarrow \mathcal{I}$.

Case 3 — Intracellular stock + continuous export (mean-linear accumulation).

$$\mathcal{I}' = \gamma T \mathcal{V} - d_{\mathcal{I}} \mathcal{I}, \quad Q' = \nu \mathcal{I} - \varepsilon Q - d_{\mathcal{I}} Q, \quad \mathcal{V}' = \varepsilon Q - c \mathcal{V}.$$

Quasi-steady stock: $p_{\text{eff}} = \varepsilon \nu / (\varepsilon + d_{\mathcal{I}})$; Case 1 is the fast-export limit. Prefer immigration-type ν for HIV-like biology (Section 35).

Case 4 — Stock + load-proportional full dump (mean-field reset). Dump flux $\propto Q^2/\mathcal{I}$ (from $\delta \mathbb{E}[X^2]$ with $X \sim$ load per cell closure $Q/\mathcal{I}$). Recovers constant- p only in mature-cell limit.

Case 5 — Eclipse + stock + proportional dump. Incidence $\rightarrow E \rightarrow \mathcal{I}$; only productive cells accumulate Q and dump.

31 Reset-catastrophe process (intermediate between budding and BDC)

[SOURCE] new_notes2/Core_Publishable_Work_Summary.tex §“Reset-catastrophe process and its renewal BMVR”.

While the cell is alive: birth βn , death μn , optional immigration ν , and **dump that resets** $n \rightarrow 0$ releasing $+n$ at rate δn . Cell dies at separate rate $d_{\mathcal{I}}$. Unlike absorbing BDC, one cell can dump many times.

$$S(a) = e^{-d_I a}, \quad g(a) = \delta e^{-d_I a} \mathbb{E}[X_a^2] \quad (47)$$

with $\mathbb{E}[X_a^2]$ for the non-absorbing reset process (linear rates $\Rightarrow$ closed moment ODEs). Mature-cell limit: $p_\infty = \delta \mathbb{E}_\pi[X^2]$. Same renewal BMVR skeleton as Result 20.1.

32 Boolean / partial-release continuum

[SOURCE] `new_notes2/Core_Publishable_Work_Summary.tex` (boolean / partial-release and batch options near end of document).

On a release event at load n , each unit released independently with probability ϕ (boolean thinning): $B \mid n \sim \text{Binomial}(n, \phi)$. Limits: $\phi \rightarrow 0$ tiny export; $\phi = 1$ full clear (Cases 4–5 / absorbing dump). Event rate free ($\delta, \delta X, \delta X^2, \dots$).

Mean-field sketch ($\lambda = \delta X$):

$$Q' = \nu \mathcal{I} - \delta \phi \frac{Q^2}{\mathcal{I}} - d_I Q, \quad \mathcal{V}' = \delta \phi \frac{Q^2}{\mathcal{I}} - c \mathcal{V}.$$

[OPEN] Flooding boundary as function of ϕ ; whether closed-form kernels survive for intermediate ϕ .

33 Self-excitation / Hawkes-type intracellular models

[SOURCE] `new_notes2/Core_Publishable_Work_Summary.tex` (Hawkes / self-excitation models and BMVR feasibility tables, later sections).

Compatible with BMVR whenever the single cell supplies (S, g) (renewal always). Finite ODE BMVR needs excitation carried by a low-dimensional Markov state (ON/OFF, mean-field R , etc.). Treat as optional extensions; not required for the core lytic BDC paper.

Part G — HIV contrast and single-cell assays

34 What Hataye et al. (2019) imply

[SOURCE] `new_notes2/Core_Publishable_Work_Summary.tex` §“Distinction with HIV”; primary literature: Hataye et al., *Cell Host & Microbe* 26:748–763, 2019.

HIV **buds**; it is not a pure load-proportional terminal catastrophe. Data support: multi-stage eclipse; short productive pulse; high silent fraction; Allee-like establishment near threshold. They reject pure memoryless constant- p from $t = 0$ and pure one-shot BDC as literal HIV.

Supported for HIV	Not supported as literal HIV
Multi-stage eclipse	Pure one-type BDC terminal catastrophe
Short productive pulse then death	Memoryless constant- p BMVR from $t = 0$
High stochasticity; many silent lineages	QSD = geometric burst as HIV law
Renewal / stage structure for free virus	Independent branching alone near threshold
Allee / synergistic establishment	

34.1 Candidate HIV stage equations (compact)

Two eclipse pathways (high/low release potential); multi-stage Erlang on the high path; division/death in eclipse ($\rho \approx \mu_E \approx 0.5/\text{day}$ in Hataye fits); short productive I with rates p_A, p_B, δ_I .

Coupling: same renewal free-virus layer with kernels $(S_{\text{HIV}}, g_{\text{HIV}})$ extracted from the stage model.

Allee incidence option near threshold:

$$f(\mathcal{V}, T) = \gamma T \mathcal{V} \cdot \frac{\mathcal{V}}{K_A + \mathcal{V}}.$$

35 HIV intracellular mean growth: linear vs exponential

[SOURCE] `new_notes2/Core_Publishable_Work_Summary.tex` §“HIV intracellular mean growth: linear vs exponential”.

HIV does not amplify infectious units by intracellular branching of virion-like particles (fixed provirus template; Pol II transcription; Gag not self-replicating). Recommended coarse regime:

$$\text{eclipse} \longrightarrow \text{mean-linear (immigration-type) accumulation of } X. \quad (48)$$

Tat positive feedback raises the production rate up to a late ceiling; it does not systematically overshoot the fully-ON rate. Pure $m' = rm$ BDC is a poor literal HIV intracellular model (fine for phage genome races / intracellular bacteria).

36 Single-cell assay predictions (lytic BDC)

[SOURCE] `new_notes/BMVR_extension_notes.tex` Part II §“Confrontation with single-cell data”; `new_notes2/Core_Publishable_Work_Summary.tex` §“Single-cell assay predictions (lytic BDC)”.

For genuinely lytic systems (*Y. pestis*, *F. tularensis*, lytic phage), limiting-dilution / Hataye-type assays:

1. fraction b of wells show no release ever;
2. cumulative release is a step: zero, then one terminal jump;
3. jump times $\sim \delta J(t)/(1 - b)$;
4. jump sizes geometric with ratio $1/a$ (same as QSD);
5. between-well variance = $\text{Var}(W_\infty)$ from (15).

Too strong for HIV as literally stated; use for lytic hosts, or as the bursting endpoint of the spectrum.

Part H — Two-type process and eclipse

37 GATE as a mechanistic eclipse

[SOURCE] Chapter 5 of the thesis (outside this folder); tasks and GATE observation in `new_notes/BMVR_extension_notes.tex` Part II §“Two-type extension and the eclipse phase”, Part III Tasks (two-type moments / D); summary in `new_notes2/Core_Publishable_Work_Summary.tex` §“Two-type process and the eclipse phase”.

Two-type BDC with conversion rate ν and type-specific catastrophe rates (δ_1, δ_2) . Backward system:

$$S' = \lambda_1 S^2 - (\lambda_1 + \mu_1 + \nu + \delta_1)S + \mu_1 + \nu G, \quad (49)$$

$$G' = \lambda_2 G^2 - (\lambda_2 + \mu_2 + \delta_2)G + \mu_2, \quad S(0) = G(0) = 1. \quad (50)$$

Remark 37.1 ([NEW] observation). Regime $\delta_1 = 0 < \delta_2$ (no burst before conversion): $S'(0) = 0$ — zero initial hazard, then rising. That is the eclipse-phase signature usually inserted by hand; here it emerges mechanistically with duration set by ν .

[OPEN] Need for two-type renewal BMVR:

- release kernel $g = \delta_1 \mathbb{E}[X^2 \mathbf{1}_{\tau_c > a}] + (\delta_1 + \delta_2) \mathbb{E}[XY \mathbf{1}] + \delta_2 \mathbb{E}[Y^2 \mathbf{1}]$;
- analogue of D (extinction without catastrophe) and two-type $\hat{I}$.

Moment hierarchy does not close upward; differentiate the backward system (analogue of obtaining K from the Riccati in one type).

Part I — Verification summary and condensed errata

38 Independent numerical checks performed

Parameters used repeatedly: $(1, 0.2, 0.05)$, $(1, 0, 0.1)$, $(1, 0.9, 0.1)$, $(1, 0.5, 1/3)$.

Identity	Result
Vieta / $AB = \delta/\beta$ / θ	exact
$\hat{I} = I - D$; Riccati for I, D	exact
$J = -I'/\delta$; $V' = \delta K$	exact
V_∞ forms; burst mean $\sum k(\delta/\beta)a^{-k}$	exact
$\int \hat{I} = \beta^{-1} \log(a/(a-1))$	trapz error $\sim 10^{-9}$
Flooding difference formula; $G(z_{\text{ext}}) = z_{\text{ext}}$	machine precision
$L > 1 \Leftrightarrow 1/\delta < 1/\beta + 1/\mu$	exact (quadratic test)
$L = 1 \Rightarrow G_{\text{off}}$ geometric	exact
MOI K_k vs master equation	match
$V_\infty^{(k)}$ integral formula vs closed form	match
$\mu = 0$: $V_\infty^{(k)} = k + \beta/\delta$	exact
$p_{\text{eff}}(r)$ decreasing; limits $r = 0, \infty$	numerical
$r_{\text{bud}} > r_{\text{burst}}$ at matched R_0 , lifetime	numerical
<i>Result 20.1 (renewal BMVR) — suite <code>verify_result_20_1.py</code></i>	
Kernel inputs to renewal system	5/5 PASS
$\gamma T = 0$ reduction to single-cell convolutions	15/15 PASS
V -equation residual under reinfection	5/5 PASS
Exponential kernels $\equiv$ classical BMVR	3/3 PASS (err. $< 5 \times 10^{-5}$)
Asymptotic growth = r_* of char. eq.	5/5 PASS (err. $\sim 10^{-7}$)
R_0 threshold scan	8/8 PASS
$p_{\text{eff}}(r_*)$ classical shares r_*	4/4 PASS
Gillespie cohort vs renewal ($\gamma T = 0$)	2/2 PASS (rel. $L^2 \sim 1\text{--}2\%$)
V_0 seed, lifetime output, positivity	7/7 PASS
Result 20.1 total	53/53 PASS

See Section 21 for full tables, figures, and verdict.

39 Condensed errata (must-fix before thesis submission)

[SOURCE] full tagged catalogue: `new_notes/BMVR_extension_notes.tex` Part IV (Chapters 1, 3, 4, 5, 6). Paths below are the files to edit.

Chapter 1. Infection terms must be driven by V , not I . Fix notation collisions $(\beta, \delta, \lambda, I, V)$ before joining intracellular and population models. (Thesis intro folder, outside this directory; see extension-notes Part IV.)

Chapter 3 — major. Files under 3 BDC `core/document MAIN/sections/`.

1. $\hat{I}$ formula and ODE wrong for $\mu > 0$; use Result 6.2. (`05_analytical_quantities.tex`, `06_derivation_....tex`)
2. $\hat{I}_k = \hat{I}^k$ wrong; use $I^k - D^k$ (`05_analytical_quantities.tex`).
3. $\mathbb{E}[W^2]$ signs and $+1$; use (15) (`08_variance_and_standard_deviation_of.tex`).

4. $K(I)$: keep §11 form / (12); discard §9 polynomial (09_vs 11_variable_dependencies_on.tex).
5. Delete duplicated section pairs 08/10 and 09/11.
6. Unify $\lambda \rightarrow \beta$, $Y_t \rightarrow W_t$, $H \rightarrow R$, $\mathcal{K}$ vs K .

Chapter 4 — major. Files under document MAIN/sections/.

1. §9 $p = \langle X \rangle_{\text{QS}}$ is irreparable as written; replace by renewal BMVR (Part D) (09_potential_application_in_07_burst_size_distribution.tex).
2. State QSD = geometric($1/a$); prove burst law analytically (done) (06_quasi_stationary_distribution.tex).
3. Resolve “circularity” with consistent conditioning (done) (08_further_considerations.tex).
4. Write $\varphi = \delta J$ wherever “known pdf” is mentioned.

Earlier notes that this document supersedes on corrections.

1. new_notes2/Core_Publishable_Work_Summary.tex: MOI g_k lacked I -powers — use Part C here.
2. Distinguish ODE R_0 from branching $m = qV_\infty$ in any write-up.
3. Prefer this file (new_notes3/) as the corrected master.

Part J — Formula tables and roadmaps

40 Master formula table (single cell)

Quantity	Closed form
a, b	$\eta \pm \sqrt{\eta^2 - \mu/\beta}, \eta = (\beta + \mu + \delta)/(2\beta)$
A, B, θ	$a - 1, 1 - b, \beta(a - b)$
$I(t)$	$(aB + bAw)/(B + Aw)$
$D(t)$	$ab(w - 1)/(aw - b)$
$\hat{I}(t)$	$(a - b)^2 w / [(B + Aw)(aw - b)]$ [CORRECTED]
$\hat{I}_k$	$I^k - D^k$ [CORRECTED]
$J(t)$	$(a - b)^2 w / (B + Aw)^2$
$K(t)$	$[1 + 2\beta(1 - I)/\delta]J$
$V(t)$	$(1 - I)[1 + \beta(1 - I)/\delta]$
V_∞	$a(1 - b)/(a - 1)$
$\mathbb{E}[\mathcal{K} \mid \text{burst}]$	$a/(a - 1)$
$\mathbb{E}[W_t^2]$	(15) [CORRECTED]
$\varphi(t)$	$\delta J(t), \text{ mass } 1 - b$
$\mathbb{P}\{\mathcal{K} = k\}$	$(\delta/\beta)a^{-k} \ (k \geq 1)$
$\text{QSD}(n)$	$(a - 1)a^{-n}$ [NEW]
$\mathbb{E}[T_{\text{prod}}]$	$\beta^{-1} \log(a/(a - 1))$ [NEW]
g_k (MOI)	$\delta[kKI^{k-1} + k(k - 1)J^2I^{k-2}]$ [CORRECTED]
$V_\infty^{(k)} \ (\mu = 0)$	$k + \beta/\delta$ [NEW]

41 Master formula table (population)

Quantity	Formula
Renewal system	$\mathcal{I} = \mathcal{I}_0 \hat{I} + i * \hat{I}, \mathcal{V}' = \mathcal{I}_0 g + i * g - c\mathcal{V}$
g	δK
$p_{\text{eff}}(r)$	$\delta \widetilde{K} / \widetilde{\hat{I}}$
$d_{\text{eff}}(r)$	$1/\widetilde{\hat{I}} - r$
Char. eq.	$r + c = \gamma T \delta \widetilde{K}(r)$
R_0^{ODE}	$\gamma T V_{\infty} / c$
$G_{\text{off}}(z)$	$b + (\delta/\beta) y / (a - y), y = 1 - q + qz$
$z_{\text{ext}}^{\text{burst}}$	$[a(1 - q + qb) - 1 + q]/q$
Flooding criterion	$1/\delta < 1/\beta + 1/\mu \ (L > 1)$
$L = 1$	offspring law = geometric [NEW]
r trade-off	$r_{\text{bud}} > r_{\text{burst}}$ at matched means [NEW]

42 Suggested thesis-chapter structure

1. Introduction — BMVR's constant-release assumption; lytic vs budding; research questions.
2. Single-cell BDC — definition; roots; $I, D, \hat{I}$ (corrected); moments; V simplified; $\mathbb{E}[W^2]$ corrected.
3. Burst time, size, QSD — identity burst = QSD; size-biasing; lifetime.
4. MOI and chained transfer — corrected kernels; $N = 2$ size laws.
5. Renewal BMVR — why §9 fails; kernels; equations; overlays.
6. Effective parameters — $p_{\text{eff}}(r)$; R_0 ; identifiability.
7. Flooding and trade-offs — main theorem; $L = 1$ law match; r comparison.
8. Spectrum — Cases 1–5; reset; boolean partial release (brief).
9. HIV contrast — what not to claim; Hataye-aligned stage model (brief).
10. Discussion — single-cell predictions; open tasks; literature.

43 Suggested paper structure (core publishable unit)

1. Intro — constant p vs lysis; closed-form kernels as contribution.
2. Single-cell results — compact catalogue + burst = QSD.
3. Renewal embedding — equations; $p_{\text{eff}}(r)$; R_0 invariant.
4. Flooding theorem — criterion; $L = 1$ geometric coincidence; variance.
5. Growth-rate trade-off — $r_{\text{bud}} > r_{\text{burst}}$.
6. Numerics — kernel plots; NEW vs CLASSICAL overlays.

7. Discussion — fitting prediction (acute vs set-point p); assay predictions for lytic systems; HIV caveat; open two-type eclipse.

Do not claim: novelty of age-structure itself; pure BDC as HIV mechanism; recovery of (β, μ, δ) from p alone.

44 Ordered open tasks

Task 44.1 (Literature positioning). Carruthers (*F. tularensis*); Pearson–Krapivsky–Perelson; Nelson–Gilchrist–Perelson; Gilchrist–Coombs nested models; Heffernan–Wahl burst size; BMVR attribution (Nowak–Bangham 1996; Perelson et al. 1996 vs McLean 1993).

Task 44.2 (Conceptual proof of QSD = burst size).

Task 44.3 (Two-type killed second moments and D). Main obstacle to eclipse-aware renewal BMVR from Ch.5.

Task 44.4 (Flooding boundary vs real pathogen parameters). Map $\delta(\beta + \mu) > \beta\mu$ for *Y. pestis* in macrophages; size of advantage via (42) style difference.

Task 44.5 (Figure: $p_{\text{eff}}(r)$ and $d_{\text{eff}}(r)$ over physiological r). Sells the “ p is not a constant of the pathogen” claim.

Task 44.6 (Population-level variance from $\text{Var}(W)$).

Task 44.7 (Partial-release continuum and flooding in ϕ).

Task 44.8 (Logistic intracellular birth — sensitivity of kernels). Referee defence: load-proportional burst already regulates; QSD is the conditional stabilisation.

45 One-page story (for introductions)

Many within-host models treat infected cells as black boxes releasing virions at constant rate p (BMVR). Lytic pathogens grow inside a cell and dump their load in a single burst. Model one cell as a classical BDC process; derive closed forms for survival, load, release, burst timing and size; embed those formulas in a renewal BMVR so the population model is forced by the intracellular process.

At fixed total output per cell, R_0 is unchanged by bursting, but generation times change. Bursting lowers establishment extinction relative to matched-mean budding if and only if $1/\delta < 1/\beta + 1/\mu$ (always when $\mu = 0$); at the boundary the offspring laws coincide. Deterministic growth is typically faster under budding. Effective release rate p_{eff} falls as epidemic growth rate rises (young cells dominate). Three intracellular rates cannot be recovered from two BMVR parameters alone.

The cleanest single-cell fact: the burst-size distribution is exactly the quasi-stationary load distribution — both geometric with ratio $1/a$.

A Quick reference: common specialisations

$\mu = 0$. $b = 0$, $a = 1 + \delta/\beta$, $D \equiv 0$, $\hat{I} = I$, $V_\infty = 1 + \beta/\delta = \langle X \rangle_{\text{QS}}$, $\mathbb{E}[T_{\text{prod}}] = \beta^{-1} \log(1 + \beta/\delta)$, $V_\infty^{(k)} = k + \beta/\delta$, flooding always favours burst for establishment, $\mathbb{P}\{\mathcal{K} = k\} = s\rho^{k-1}$ with $s = \delta/(\beta + \delta)$, $\rho = 1 - s$.

$L = 1$ boundary. $\delta = \beta\mu/(\beta + \mu)$, $b = \delta/\beta$, offspring law geometric, $z_{\text{ext}}^{\text{burst}} = z_{\text{ext}}^{\text{bud}}$, equal variances.

Young-cell / fast-growth limit. $p_{\text{eff}} \rightarrow \delta$, $d_{\text{eff}} \rightarrow \mu + \delta$.

Set-point / $r = 0$ matching. $p_{\text{eff}} = V_\infty/\mathbb{E}[T_{\text{prod}}]$, $d_{\text{eff}} = 1/\mathbb{E}[T_{\text{prod}}]$.